Mini Project Report on Text Categorization Using Natural Language Processing

Group Members

- 231636 – Mohd. Hussain
- 231257 – Mohd. Zaid
- 221232 – Mohd. Ikhlas Husain

Experiment 1: Study of NLP Applications and Problem Statement Formulation

1 Introduction

Text Categorization, also known as Text Classification, is one of the most important tasks in Natural Language Processing (NLP). It is the process of automatically assigning one or more predefined categories or labels to text documents based on their content. With the rapid growth of digital information, organizations receive millions of emails, documents, news articles, reviews, and social media posts every day. Manually organizing this huge amount of textual data is difficult, time-consuming, and expensive. Text categorization provides an automated solution by classifying documents into appropriate categories. It is widely used in email spam filtering, sentiment analysis, document organization, topic detection, fake news detection, customer support systems, recommendation systems, and many other real-world applications.

2 Various Applications of NLP

1. **Machine Translation:** Converts text from one language to another while preserving meaning.
2. **Chatbots and Virtual Assistants:** Provide automated responses to user queries in customer support and personal assistance systems.
3. **Sentiment Analysis:** Detects the emotional tone of text such as positive, negative, or neutral.
4. **Spam Detection:** Identifies unwanted or malicious content in emails and messages.
5. **Text Categorization:** Produces a shorter version of a document while keeping the important information.
6. **Information Retrieval:** Helps search engines and document systems retrieve relevant content quickly.

3 Selected Topic: Text Catogarization

Text Categorization is the process of automatically assigning predefined labels or categories to text documents using machine learning, deep learning, or rule-based techniques.

For example, • News Article → Sports ,• Email → Spam, • Product Review → Positive, • Research Paper → Artificial Intelligence

4 Problem Statement

The rapid growth of digital text from emails, social media, news articles, and websites has made it difficult to organize and analyse information manually. Traditional methods are time-consuming, error-prone, and cannot efficiently handle large volumes of text. Therefore, an automated text categorization system is needed to classify documents into predefined categories accurately and quickly, improving information retrieval, search, and decision-making.

5 Objective of the Mini Project

- To study common real-world applications of NLP.
- To formulate a practical problem statement based on text Categorization.
- To design a simple NLP-based application that Categorizes long text documents.
- To allow users to upload or paste text and obtain a concise summary.
- To explore related features such as sentiment analysis and word cloud generation.

6 Objective of the Mini Project

- To study common real-world applications of NLP.
- To formulate a practical problem statement based on text Categorization.
- To design a simple NLP-based application that Categorizes long text documents.
- To allow users to upload or paste text and obtain a concise summary.
- To explore related features such as sentiment analysis and word cloud generation.

7 Objective of the Mini Project

- To study common real-world applications of NLP.
- To formulate a practical problem statement based on text Categorization.
- To design a simple NLP-based application that Categorizes long text documents.
- To allow users to upload or paste text and obtain a concise summary.
- To explore related features such as sentiment analysis and word cloud generation.

8 Objective of the Mini Project

- To study common real-world applications of NLP.
- To formulate a practical problem statement based on text Categorization.
- To design a simple NLP-based application that Categorizes long text documents.
- To allow users to upload or paste text and obtain a concise summary.
- To explore related features such as sentiment analysis and word cloud generation.

9 Scope of the Project

The mini project focuses on automatic extractive text categorization using Python and NLP libraries. The system can accept plain text and PDF input, extract the relevant content, and generate a summary by identifying the most important sentences. The project is suitable for academic documents, online articles, and general text inputs.

10 Methodology

The application follows a simple workflow:

1. Data Collection – Gather text documents.
2. Text Preprocessing – Clean and prepare the text.
3. Feature Extraction – Convert text into numerical features.
4. Model Selection – Choose a classification algorithm.
5. Model Training – Train the model using labelled data.
6. Model Evaluation – Test the model's performance.
7. Text Classification – Assign categories to new text.

11 Project Screenshots

The following screenshots show the actual working of the mini project. The first image is the input screen, and the second image is the generated output screen.

```
# 1. Custom Tweet
tweet_input = "Having a great day with friends, drinking coffee and enjoying the weather!"

# 2. Clean Text
```

Figure 1: Input Screen of Text Categorize

```
==========================================================
📥 INPUT TWEET: Having a great day with friends, drinking coffee and enjoying the weather!
==========================================================
✅ RESULT: NORMAL TWEET
📊 DISASTER PROBABILITY: 23.58%
==========================================================
```

Figure 2: Output Screen Showing Categorized Text


```
# 1. Custom Tweet
tweet_input = "Having a great day with friends, drinking coffee and enjoying the weather!"

# 2. Clean Text
```


Figure 2: Input Screen of Text Categorize

```
print( = * 55)

==========================================================
📥 INPUT TWEET: Huge explosion and severe fire broke out near the highway!
==========================================================
🚨 RESULT: DISASTER TWEET
📊 CONFIDENCE: 99.59%
==========================================================
```

Figure 4: Output Screen Showing Categorized Text

```
messag clean may involv thing like adjac space tab
```

```
[*]: #FINAL PREPROCESSING
def finalpreprocess(string):
    return lemmatizer(stopword(preprocess(string)))

df_train['clean_text'] = df_train['text'].apply(lambda x: finalpreprocess(x))
df_train=df_train.drop(columns=['word_count','char_count','unique_word_count'])
df_train.head()
```

Word2Vec model

Figure 5: Data model of Text Categorizer

12 Tools and Technologies Used

Component	Technology Used
Programming Language	Python
IDE	Jupyter Notebook
NLP Libraries	NLTK, Scikit-learn
Data Handling	Pandas, NumPy
ML Algorithms	Naïve Bayes, SVM
Visualization	Matplotlib
Dataset	CSV/Text Dataset

13 Expected Outcome

The expected outcome of the project is a user-friendly web application that can Categorize long documents into shorter versions. This helps users quickly understand the main idea of a text, improves productivity, and supports efficient document review.

14 Applications in Real World

Text Categorization is used in news aggregation platforms, research article preview systems, legal document analysis, customer feedback processing, and educational tools. It is especially useful where large amounts of text must be understood in limited time.

15 Conclusion

Text Categorization is an essential task in Natural Language Processing that automatically classifies textual documents into predefined categories. It helps organizations efficiently manage large volumes of text, improves search and retrieval, enables automation, and supports intelligent decision-making. Traditional machine learning methods such as Naive Bayes and SVM are widely used, while modern transformer-based models like BERT and GPT have significantly improved classification accuracy. As NLP continues to evolve, text categorization will become even more accurate, scalable, and useful across healthcare, education, finance, business, cybersecurity, and many other domains.

References

- Aggarwal, C. C., & Zhai, C. (2012). Mining Text Data. Springer. https://doi.org/10.1007/978-1-4614-3223-4
- Sebastiani, F. (2002). Machine Learning in Automated Text Categorization. ACM Computing Surveys, 34(1), 1–47. https://doi.org/10.1145/505282.505283
- Flask Documentation: https://flask.palletsprojects.com/